

A Review of Attention-Deficit/Hyperactivity Disorder Complicated by Symptoms of Oppositional Defiant Disorder or Conduct Disorder

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ABSTRACT: *Background:* Attention-deficit/hyperactivity disorder (ADHD) is a highly prevalent disorder with significant functional impairment. ADHD is frequently complicated by oppositional symptoms, which are difficult to separate from comorbidity with oppositional defiant disorder, conduct disorder, and aggressive symptoms. This review addresses the impact of oppositional symptoms on ADHD, disease course, functional impairment, clinical management, and treatment response. *Review of clinical evidence:* Oppositional defiant disorder or conduct disorder may be comorbid in more than half of ADHD cases and are more common with the combined than with the inattentive ADHD subtype. Comorbid symptoms of oppositional defiant disorder and conduct disorder in patients with ADHD can have a significant impact on the course and prognosis for these patients and may lead to differential treatment response to both behavioral and pharmacologic treatments. *Impact on clinical management:* Assessment of oppositional symptoms is an essential part of ADHD screening and diagnosis and should include parental, as well as educator, input. Although clinical evidence remains limited, some stimulant and nonstimulant medications have shown effectiveness in treating both core ADHD symptoms and oppositional symptoms. *Conclusions:* Oppositional symptoms are a key consideration in ADHD management, although the optimum approach to treating ADHD complicated by such symptoms remains unclear. Future research should focus on the efficacy and safety of various behavioral and medication regimens, as well as longitudinal studies to further clarify the relationships between ADHD, oppositional defiant disorder, and conduct disorder.

(*J Dev Behav Pediatr* 31:427–440, 2010) **Index terms:** attention-deficit/hyperactivity disorder, oppositional defiant disorder, comorbidity, conduct disorder.

Attention-deficit/hyperactivity disorder (ADHD) is a common developmental neuropsychiatric disorder broadly characterized by hyperactivity, inattention, and/or impulsivity.^{1–3} Three subtypes of ADHD are defined by the DSM-IV-TR: combined type, predominantly inattentive type, and predominantly hyperactive-impulsive type.¹

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ADHD is associated with significant impairment in social, family, financial/economic, educational, and vocational functioning, as well as with substantial costs to society in both financial terms and adverse outcomes.^{1,2,4,5} It is also highly prevalent, with a lifetime prevalence among children aged 4 to 17 years estimated at 7.8% in the National Survey of Children's Health.⁵

There is new and emerging evidence to suggest that “complex” ADHD (i.e., ADHD accompanied by other disorders) is the rule, rather than the exception, in clinical treatment settings. Oppositional defiant disorder is the most common psychiatric comorbidity in children with ADHD, occurring in ~60% of patients. Another disruptive behavior disorder, comorbid conduct disorder, occurs in ~20% of patients with ADHD.⁴ These comorbidities complicate ADHD treatment by altering the course, prognosis, and types of impairment seen. They also have been associated with family risk factors (cumulative family stress, dysfunctional parenting, and parental psychopathology) and have implications for treatment.^{6,7} Therefore, all clinicians who treat ADHD, including pediatricians, developmental-behavioral pediatricians, and pediatric psychologists, need to become more aware of the presence and complications of Oppositional defiant disorder or conduct disorder symptoms as they present in children with ADHD as well as

the implications of such symptoms in determining appropriate treatment. Presently, there are new, published phenomenology investigations and psychopharmacology clinical trials that begin to address these issues.

This review addresses clinical features and risk factors of Oppositional defiant disorder and oppositional symptoms and conduct disorder, first alone, and then as comorbidities to ADHD. The review then turns to the impact of Oppositional defiant disorder/conduct disorder symptoms on clinical decision making during diagnosis and treatment of patients with ADHD.

SYMPTOMS OF OPPOSITIONAL DEFIANT DISORDER AND CONDUCT DISORDER

DSM-IV-TR groups Oppositional defiant disorder (ODD) and conduct disorder (CD) with attention-deficit/hyperactivity disorder (ADHD) under Attention-Deficit and Disruptive Behavior Disorders.¹ The hallmark of ODD is “a recurrent pattern of negativistic, defiant, disobedient, and hostile behavior directed at authority figures that persists for at least six months.”¹ ODD is frequently manifested as defiance of authority, excessive anger and resentment, bullying, frequent loss of temper, spitefulness/vindictiveness, and blaming others. The onset of ODD is typically before 8 years of age and only rarely after early adolescence.^{1,2} ODD is characterized by the presence of at least 4 of 8 oppositional symptoms, lasting 6 months or more, and accompanied by significant impairment.¹ However, a patient can have 3 or fewer oppositional symptoms, have symptoms for <6 months, or have symptoms without impairment. In any of these cases, the patient may have oppositional symptoms without having the diagnosis of ODD.

The essential feature of CD is “a repetitive and persistent pattern of behavior in which the basic rights of others or major age-appropriate societal norms or rules are violated.”¹ CD is manifested in overt aggression toward people or animals, property destruction, theft and deceitfulness, and serious violations of rules and laws. If at least one criterion behavior is present before 10 years of age, CD is classified as childhood-onset type; otherwise, it is classified as adolescent-onset type. Childhood-onset CD is associated with greater severity, worse prognosis, and, frequently, persistence into adulthood as antisocial personality disorder.¹

DSM-IV established a hierarchical relationship between CD and ODD such that CD is treated as a more severe or progressed form of behavior disorder on the same continuum as ODD. If CD is diagnosed, ODD is not.¹ The assumption that CD encompasses ODD is based on the relatively greater seriousness and impairment that is associated with CD, and on research that implies a developmental progression.⁸ When studied cross sectionally, cases of CD frequently exhibit many of the criterion behaviors of ODD. Studied retrospectively, most cases of early-onset CD presented as ODD at an early age and progressed to CD.¹ Prospectively, ODD appears to be an antecedent to childhood-onset CD,

although the majority of patients with ODD do not go on to develop CD.¹

The similarities between ODD and CD have prompted discussion regarding whether they are truly separate disorders or different manifestations of the same disorder.^{9,10} According to Reiff and Stein, broadly speaking, oppositional behaviors of mild to moderate intensity are more likely to be components of ADHD, whereas those of moderate to severe intensity are more likely to represent ODD/CD.² Over time, changes in symptoms of ADHD, ODD, and CD are all correlated.^{11,12} Even when considered as separate entities, ODD, CD, ADHD, and aggression are commonly comorbid and have often been studied together.^{9,12,13} Many studies have compared patients with ADHD to patients with ADHD and ODD or CD without distinguishing between the 2 comorbidities. A further complication is that younger children with ODD who will eventually develop CD cannot be reliably distinguished from those who will not, although the early appearance of persistent and variegated conduct problems before age 10 years (subclinical CD) is a strong predictor of later meeting diagnostic criteria for CD.¹⁴ Therefore, some confounding of correlates specific to ODD versus CD is inevitable.

RISK FACTORS FOR OPPOSITIONAL DEFIANT DISORDER AND CONDUCT DISORDER

Socioeconomic and Family Factors

Risk factors that have been linked to the development of Oppositional defiant disorder (ODD) and conduct disorder (CD) include poverty, parental psychopathology, coercive family process, lack of parental monitoring, community violence, living in public housing, and poor and disadvantaged neighborhoods.^{7,15–17} Parental psychopathology and socioeconomic factors may not be directly amenable to intervention, but targeting intermediate factors such as parental behavior (coercive family process, lack of monitoring/supervision, poor parent-child interactions and parenting skills, lack of monitoring of association with antisocial peers, and truancy) and increasing opportunities for positive socialization (organized activities, skill-building activities, and exposure to positive role models) are more practical and may have protective effects.

Early and Persistent Physical Aggression

Longitudinal studies have shown that many children exhibit physical aggression during toddlerhood and early childhood but in most cases, these behaviors decline in frequency as the result of preschool and school-aged socialization.^{18–21} In boys, physical fighting during the preschool years and reactive aggression (aggression in response to frustration, impulsivity, or fear of threat) are common and do not identify those boys who will eventually develop persistent conduct problems.¹⁶ A minority of children (about 7–15%) follow a pattern of elevated aggression, beginning with higher-than-normal levels of physical aggression during early childhood. Physical fighting that persists beyond early childhood, and pro-

active aggression (nonimpulsive goal-directed aggression that is not an immediate response to provocation and is driven by reward contingencies), have been associated with a different developmental trajectory with progressive acquisition of more disruptive behaviors.^{16,22,23} Children in this trajectory not only demonstrate persistent physical aggression but also increase the severity of their aggression and acquire additional antisocial behaviors by adolescence.^{22,23} Early child- and family-focused intervention has been recommended as a means of possibly changing the pernicious natural course of this trajectory.

Extension to Nonparental Adults

Normatively, young children express aggression most often toward parents and siblings rather than authority figures, such as teachers, outside the home. A potential sign that a child is progressing in an antisocial trajectory is when adults who are not the child's parent are among the recipients of oppositional and aggressive behavior.

Attention-Deficit/Hyperactivity Disorder as a Risk Factor for ODD Conduct Disorder

The greater prevalence of ODD/CD in children with ADHD, compared with the general population, suggests that having ADHD increases the odds of developing ODD/CD (discussed below). This is borne out in longitudinal studies. The co-occurrence of ADHD and childhood-onset CD may pose far more risk than either disorder alone. These disorders may synergize in the direction of earlier onset, greater persistence, and greater severity, with an especially poor longitudinal prognosis.^{16,24}

PREVALENCE OF OPPOSITIONAL DEFIANT DISORDER OR CONDUCT DISORDER SYMPTOMS IN ATTENTION-DEFICIT/HYPERACTIVITY DISORDER

Although the prevalence of oppositional symptoms in Attention-deficit/hyperactivity disorder (ADHD) has not been determined, studies have evaluated the prevalence of the comorbid diagnoses of Oppositional defiant disorder and conduct disorder in patients with ADHD. A survey of pre-1990 evidence found that comorbid Oppositional defiant disorder or conduct disorder ranged from 15% to 65% in patients aged 6 to 17 years with ADHD, with most reports suggesting an incidence of ~50%.^{25,26} A more recent analysis estimated the comorbidity of Oppositional defiant disorder with ADHD at ~60% and the comorbidity of conduct disorder with ADHD at ~15%.⁴ The prevalence of Oppositional defiant disorder/conduct disorder appears greater in patients with the combined ADHD subtype than in those with the inattentive subtype, possibly due in some part to greater executive dysfunction and difficulties with impulsivity and response inhibition observed in patients with the combined subtype.²⁷

IMPACT OF COMORBID OPPOSITIONAL DEFIANT DISORDER/CONDUCT DISORDER SYMPTOMS ON PATIENTS WITH ATTENTION-DEFICIT/HYPERACTIVITY DISORDER

Although several studies have examined the impact of comorbid Oppositional defiant disorder (ODD), conduct disorder (CD), or both on children and adolescents with Attention-deficit/hyperactivity disorder (ADHD), fewer have examined the impact of subthreshold oppositional symptoms. Both types of studies, however, have shown that the presence of comorbid ODD or CD symptoms in patients with ADHD can negatively affect the course and prognosis of ADHD itself, subsequent behaviors and psychiatric diagnoses, family dynamics, and educational outcomes.⁷

ADHD comorbidity with ODD or CD may be associated with increased symptom severity of both learning problem/disorders and disruptive behavior disorders. A cross-sectional study of 200 boys with ADHD (mean age of 10.6 years) revealed increasing ADHD symptom severity (using the parent-completed ADHD Rating Scale Version IV) as well as increasing aggression when comparing ADHD alone with ADHD + ODD and with ADHD + CD ($p < .001$ for all pairwise comparisons). This study also demonstrated that overt aggression (including property destruction and assaults on others and self) was highly associated with ADHD + ODD compared with ADHD alone, despite the absence of aggression as a criterion symptom for ODD.¹⁰ In addition, ADHD treatment history was strongly influenced by oppositional symptoms: boys with ADHD + CD or ADHD + ODD were significantly more likely to have received ADHD medications than were those with "pure" ADHD ($p < .001$).¹⁰

Comorbid ODD/CD may affect parents' self-rating of their effectiveness as well as patients' educational outcomes. In one study, only 16% and 14% of parents of boys ≤ 12 years of age with ADHD+ODD and ADHD+CD, respectively, reported managing their son's behavior "very well," compared with 46% of parents of boys with ADHD alone ($p < .001$). Further, self-reported agreement between spouses with respect to child management issues was higher for parents of sons with ADHD only (73%) than for those of sons with ADHD+ODD or ADHD+CD (53 and 42%, respectively). The same study also found that younger boys with ADHD alone had a lower rate of placement in special education (59%) than did those with ADHD+ODD (76%) or ADHD+CD (89%). Among older boys, the likelihood of placement in a regular classroom was lower for those with ADHD+CD (57%) than for those with ADHD alone or ADHD+ODD (85 and 87%, respectively).¹⁰

The presence of diagnosed ODD or CD, or symptoms of these disorders, in patients with ADHD further increases the risk for a range of antisocial behaviors beyond that of ADHD alone.²⁸ Both childhood ADHD and ODD/CD are predictors of increased risk for substance

use and abuse, but the presence of CD in adolescents with persistent ADHD nearly doubles the risk of substance use behaviors relative to those with persistent ADHD uncomplicated by CD.²⁹ In addition, a 10-year follow-up study of boys with ADHD stratified by Satterfield Aggression/Defiance (A/D) scores showed that those with high A/D scores were 1.65 times more likely to have been arrested for felony offenses.³⁰ Finally, a longitudinal study showed that adolescents with ADHD received poorer parental ratings of driving behaviors than did those without ADHD and that these ratings were poorer still when ADHD was complicated by symptoms of CD and/or ODD.³¹

Co-occurrence of CD or ODD with ADHD also influences the risk for other psychiatric comorbidities. The presence of ODD or ODD+CD in children with ADHD has been associated with higher baseline rates of major depression, bipolar disorder, and multiple (2 or more) anxiety disorders than the presence of ADHD alone.²⁵ Similarly, a history of ODD (with or without CD) in adults with ADHD has been associated with increased prevalence of bipolar disorder, multiple anxiety disorders, and substance use disorder.³²

There have been efforts to distinguish the effects of comorbid CD from those of ODD. A study of boys with ADHD found that subjects with ADHD+CD were most likely to have fathers with antisocial personality disorder followed by subjects with ADHD+ODD, and finally, those with ADHD but not ODD or CD.¹⁷ This is evidence of a “dose effect,” in which the group with comorbid CD had greater association with paternal antisocial personality disorder than the group with ODD (which presumably would include some cases destined to progress to CD, but also some “benign” cases without elevated genetic predisposition to antisocial behavior). A study of children of both genders with ADHD found that children with ADHD whose biological fathers had left the family and could not be located displayed significantly more CD symptoms and experienced significantly lower socioeconomic status, than those whose families were intact.³³ Observations such as these are often attributed to the loss of a father figure who provides structure, discipline, role modeling, and financial resources. However, Piffner et al³³ speculated that the higher rate of CD symptoms may arise in part because biological fathers who leave the home are more likely to have antisocial personality disorder and more likely to pass on genetic risk for conduct problems. Notably, father departure followed by repartnering and restoration of a father in the family was not accompanied by any lowering of the rate of comorbid CD. Conversely, in the well-known Isle of Kauai study factors that were associated with positive outcomes, despite risk factors, included an internal locus of control, affectional ties that provided emotional support, and support systems outside the families.³⁴

Disruptive behavior disorders that are comorbid with ADHD are also associated with dysfunctional parenting. In a study of children with ADHD, both comorbid ODD

and comorbid CD were associated with negative and ineffective discipline on the part of either mothers or fathers. Both lack of warmth and involvement by mothers, as well as paternal antisocial personality disorder, were associated with comorbid CD.³⁵ Here again is evidence of a dose effect with progressively more dysfunctional parenting being associated with more serious disruptive behavior disorders.

Children with ADHD have been reported to have more problems and conflicts with siblings than children without ADHD. However, one study of children with ADHD found that comorbid externalizing problems (symptoms of ODD and CD) were associated with less warmth and closeness and more conflict in sibling relationships, and that it was these comorbidities rather than ADHD per se that accounted for much of the problems in sibling relationships.³⁶

Finally, self-reported health-related quality of life among children and adolescents with ADHD is also negatively affected by comorbid ODD/CD. Compared to patients with ADHD with no comorbid disorder, those with comorbid ODD/CD had significantly lower scores for psychosocial health ($p < .001$); no such difference was observed for patients with ADHD and learning disabilities.³⁷

IMPLICATIONS FOR CLINICAL MANAGEMENT

Diagnosis and Identification of Comorbid Oppositional Defiant Disorder/Conduct Disorder Symptoms

It can be difficult for health care providers to distinguish symptoms of Oppositional defiant disorder (ODD) from normative or transient oppositional behavior.⁷ Clinicians can identify symptoms of ODD by their occurrence outside the appropriate developmental stage as well as by their severity and adverse effect on functioning. Signs that symptoms may not indicate the presence of ODD but rather a time-limited issue (such as a transient reaction to a stressor) include good premorbid functioning, specificity of the impairment to one or few functional domains (home, school, or community) and the presence of a recognizable precipitating factor.

Practice guidelines for attention-deficit/hyperactivity disorder (ADHD) expressly recommend evaluation of comorbidities.³⁸ Because the diagnostic criteria from the DSM-IV for disruptive behaviors were derived by combining diagnostic reports from parents, teachers, and self-reports from youths being evaluated,^{39–41} the use of multiple informants will lead to diagnostic decisions that better correspond to DSM-IV than will sole reliance on parent report (which would be expected to under-identify cases diagnosed by a multi-informant procedure).⁴² Multi-informant assessment tools such as the Conners 3—Parent scale, the Swanson, Nolan, and Pelham-IV Teacher and Parent Rating Scale, the Conners' Parent Rating Scale—Revised: Long Version, the Child Symptom Inventory, and the Vanderbilt ADHD Diagnostic Parent Rating Scale include measures of oppositional symptoms as well as of ADHD (summarized in

Table 1. Items Related to Oppositional Symptoms From the CPRS-R:L, Conners 3—Parent, the SNAP-IV Teacher and Parent Rating Scale, Child Symptom Inventory, and the Vanderbilt ADHD Diagnostic Parent Rating Scale

CPRS-R:L—Oppositional Subscale ⁴³	
Angry and resentful	
Fights	
Argues with adults	
Loses temper	
Irritable	
Actively defies or refuses to comply with adults' requests	
Touchy or easily annoyed by others	
Blames others for his/her mistakes or misbehavior	
Deliberately does things that annoy other people	
Spiteful or vindictive	
Conners 3—Parent ⁴⁶	
Loses temper	
Argues with adults	
Actively refuses to do what adults tell him/her to do	
Annoys other people on purpose	
Blames others for his/her mistakes or misbehavior	
Is irritable and easily annoyed by others	
Is angry and resentful	
Tries to get even with people	
SNAP-IV Teacher and Parent Rating Scale ^{44,45}	
Often loses temper	
Often argues with adults	
Often actively defies or refuses adult requests or rules	
Often deliberately does things that annoy other people	
Often blames others for his/her mistakes or misbehavior	
Often is touchy or easily annoyed by others	
Often is angry and resentful	
Often is spiteful or vindictive	
Often is quarrelsome	
Often is negative, defiant, disobedient, or hostile toward authority figures	
Child Symptom Inventory ⁴⁷	
Loses temper	
Argues with adults	
Defies or refuses what you tell him/her to do	
Does things to deliberately annoy others	
Blames others for own misbehavior or mistakes	
Is touchy or easily annoyed by others	
Is angry and resentful	
Takes anger out on others or tries to get even	
Vanderbilt ADHD Diagnostic Parent Rating Scale ^{48,49}	
Argues with adults	
Loses temper	
Actively defies or refuses to comply with adults' requests or rules	
Deliberately annoys people	
Blames others for his or her mistakes or misbehavior	
Is touchy or easily annoyed by others	
Is angry or resentful	
Is spiteful and vindictive	

CPRS-R:L, Conners' Parent Rating Scale—Revised: Long Form; SNAP-IV, Swanson, Nolan, and Pelham-IV; ADHD, attention-deficit/hyperactivity disorder. Reproduced with permission of Conners.⁴³ Reproduced with permission of Conners.⁴⁶ Reproduced from Gadow⁴⁷, Vanderbilt Diagnostic Parent Rating Scale,⁴⁸ and Vanderbilt Diagnostic Parent Rating Scale Scoring Instructions.⁴⁹

Table 1).^{43–49} Scales that rate symptoms on a 4-point range of severity can be used to approximate symptoms by counting those items rated above the midpoint (e.g., 2s and 3s on scales in which item severity ranges from 0 to 3) as indicating the presence of that symptom. In addition, if a symptom is counted as present whether the parent or the teacher rates it as present, DSM-IV diagnostic procedure can then be followed in terms of determining whether criteria for the disorder is met. Children and adolescents may have difficulty rating their own ADHD symptoms, but they are in a unique position to disclose any undiscovered conduct disorder (CD) behaviors in which they have engaged.

There are several self-report measures of CD behaviors that may be used, including the Youth's Inventory-4, the Impulsive and Premeditated Aggression Scale, the Adolescent Anger Rating Scale, and the Children's Inventory of Anger.^{50–53} The Youth Self Report for 11- to 17-year olds, which is based on the Child Behavior Checklist, has self-report scales for rule breaking behaviors (delinquency) and aggression.⁵⁴ However, presently, there are few CD self-report scales with adequate psychometric properties, especially for younger children.

Disruptive comorbidities are frequently the driving force when parents seek professional help, as suggested by higher rates of treatment (both medical and psychosocial) among patients with ADHD and oppositional disorders than among those with ADHD alone.¹⁰ However, parents often focus their discussion on school performance without mentioning oppositional symptoms, even when they regard these symptoms as among their child's "most concerning behaviors."⁵⁵ Parents who do not mention oppositional symptoms such as hostility and defiance will often discuss their concerns when asked directly.

Treatment

One question that remains is whether comorbid oppositional symptoms change the efficacy of treatments targeting ADHD symptoms (differential response). Another question is whether evidence-based treatments for ADHD also affect comorbid oppositional symptoms.

Differential Response: Combined Behavioral and Pharmacotherapy

Currently, the evidence regarding differential response of ADHD complicated by oppositional symptoms to both behavioral therapy and medication remains mixed. In the landmark Multimodal Treatment Study, the response of children with ADHD+ODD was similar to that of ADHD-only children with respect to ADHD as well as oppositional symptoms. Each group demonstrated meaningful response to medication treatment (predominantly methylphenidate) alone or in combination with behavioral therapy, with generally similar effect sizes for combination therapy in each group. For example, with regard to Swanson, Nolan, and Pelham total scores, the effect size (using community treatment as the comparison) of combination therapy in children

with ADHD-only was 0.51 and the effect size in children with ADHD+ODD was 0.49. The authors concluded that the presence of ODD/CD comorbidity did not appear to interact significantly with treatment response or outcomes.⁵⁶

A similarly designed study also did not find differential treatment response between patients with ADHD only and those patients with comorbid conditions with respect to ADHD as well as oppositional symptoms. This study excluded patients with CD, although patients with ADHD+ODD were included. Compared to patients with ADHD alone, those with ADHD+ODD responded similarly to methylphenidate, methylphenidate plus multimodal psychosocial treatment, and methylphenidate plus attention-control psychosocial treatment.^{57,58}

Differential Response: Pharmacotherapy

Similar to the evidence regarding differential response to behavioral therapy and medication combined, data regarding differential response to pharmacotherapy alone are also mixed. However, the preponderance of data on this topic suggests that patients with ADHD alone respond to medication in a similar manner to those patients with ADHD and oppositional symptoms.

Regarding stimulant effects on ADHD core symptoms, a study of mixed amphetamine salts extended release that included 308 subjects with ODD, 244 of whom had comorbid ADHD, provided evidence of efficacy in patients with ADHD+ODD but not ODD alone. Although the percentage of subjects who showed improvement as measured by the Clinical Global Impressions-Severity scale were similar in subjects with ADHD+ODD as in subjects with ODD alone, that percentage was not significantly higher than the placebo group in the subset of subjects with ODD alone. However, interpretation of this study is limited by the small number of subjects who had ODD alone ($n = 64$).

The same study also evaluated the impact of mixed amphetamine salts extended release treatment on oppositional symptoms. The effects of mixed amphetamine salts extended release as measured by the ODD subscale of the Swanson, Nolan, and Pelham-IV (parent rating) were statistically significant compared with placebo ($p = .024$) across the entire intent-to-treat population (including both “pure” ODD and ADHD+ODD; $n = 297$). However, the effects of mixed amphetamine salts extended release on Swanson, Nolan, and Pelham-IV parent rating scores were not statistically significant in the “pure” ODD group. Once again, however, the small sample size of the “pure” ODD population should be noted. The most common treatment emergent adverse events reported in this study were anorexia/decreased appetite, insomnia, headache, abdominal pain, and weight loss.⁵⁹

In another study, atomoxetine (0.5, 1.2, and 1.8 mg/kg/d) improved ADHD symptoms (as measured by the ADHD Rating Scale Version IV) in children with ADHD at 1.2 and 1.8 mg/kg/d, and in children with ADHD+ODD only at 1.8 mg/kg/d. Although these data support a

potential need for higher doses of atomoxetine in the treatment of children with ADHD+ODD, effect sizes indicate that the difference in efficacy between the subgroups may be of minor clinical importance. Effect sizes on the ADHD Rating Scale Version IV for children with ADHD+ODD were as follows: placebo vs atomoxetine at 1.2 mg/kg/d = 0.49; placebo vs atomoxetine at 1.8 mg/kg/d = 0.69; and placebo vs atomoxetine at 1.2 mg/kg/d + atomoxetine at 1.8 mg/kg/d = 0.57. For children with ADHD alone, effect sizes were placebo vs atomoxetine at 1.2 mg/kg/d = 0.66; placebo vs atomoxetine at 1.8 mg/kg/d = 0.50; and placebo vs atomoxetine at 1.2 mg/kg/d + atomoxetine 1.8 = 0.56.⁶⁰ It should be noted that the 1.8 mg/kg/d dose is above the recommended upper dose for treatment of ADHD in the approved US Food and Drug Administration package insert for atomoxetine.⁶¹ Headache, rhinitis, abdominal pain, pharyngitis, and anorexia were among the most common treatment emergent adverse events reported in this study.⁶² Finally, a meta-analysis of 11 studies of the non-stimulant clonidine in the treatment of ADHD found that treatment effect size was significantly reduced in patients with ADHD plus at least one comorbidity compared with ADHD alone ($p < .0001$).⁶³

In summary, these studies suggest that ADHD medications still demonstrate effectiveness for ADHD symptoms when ODD is also present. However, there is a suggestion that the effectiveness of some medications may be diminished in patients with ADHD and oppositional symptoms as compared to those with ADHD symptoms alone. One potential impact of comorbid disruptive behavior disorders that should be considered is their effect on adherence to prescribed medication regimens. One study found that the presence of ODD symptoms at school was associated with an 11-fold increase in the risk of medication discontinuation; however, parent-identified ODD was not significant as a predictor of discontinuation risk.⁶⁴

Efficacy: Behavioral Therapy

Behavioral therapy represents important and appropriate first-line treatment for patients with ADHD. Behavioral therapy aims to provide structure and incentives for appropriate behavior, give consistent praise for prosocial behaviors, and set clear and consistent goals and expectations for the child with ADHD. These strategies are generally more effective in children with ADHD than insight-driven psychosocial strategies.⁶⁵

Behavioral therapy is directed at the entire family, beginning with intensive parent management training to: educate parents on the nature of ADHD; instruct them in positive and negative reinforcement techniques; emphasize the need for consistent, realistic approaches; and plan for long-term maintenance and relapse prevention. Table 2 lists examples of techniques commonly taught to parents as part of parent management training for the management of ADHD.⁶⁶⁻⁶⁹

Similarly, the efficacy of evidence-based psychosocial treatments for disruptive behavior disorders is reason-

Table 2. Behavioral Therapy Techniques Included in PMT for the Management of ADHD^{66–69}

Technique	Description
Coercive family process	Explain to parents how parent and child reinforce oppositional behaviors in the home by negative reinforcement (becoming oppositional/aggressive to avoid aversive situations)
Positive reinforcement	Providing rewards or privileges contingent on patient's performance of desirable behavior
Balance emotional valence in the home	Help parents to express positive emotions toward the child at least in equal daily proportion to expressed negative/command communications
Time out	Denying access to rewards/privileges contingent on performance of unwanted or undesired behavior
Response cost	Withdrawing rewards/privileges contingent on performance of unwanted or undesired behavior
Token economy	Combining positive reinforcement and response cost; patient earns "net" (rewards for positive behavior minus costs for negative behavior) in form of prize
Consistency of response	Responses to the child's hostile, oppositional behaviors should be consistent across home (both parents responding in the same manner), school (teacher responds as parents to the child), and across time (in the same manner each day)
Help parents identify priorities for treatment	Help parents identify 2 or 3 core childhood behaviors to target and what behaviors they wish to ignore

PMT, parent management training; ADHD, attention-deficit/hyperactivity disorder. Reproduced from American Academy of Pediatrics.⁶⁶

ably well established. In particular, there is support for parent training for young children and skills-based training for older children, with estimated effect sizes favoring these types of treatments generally over 0.20.⁷⁰ In addition, programs such as head start may be effective as preventive interventions.⁷

Despite evidence for the effectiveness of behavior therapy in ADHD and ODD, there is little evidence regarding how (and if) these approaches should be modified when ADHD is comorbid with ODD or CD or symptoms of these disorders.⁶⁶ The active engagement of parents, via parent management training, would appear to be especially important in such cases, as the co-occurrence of ODD and CD in children with ADHD is strongly correlated with family attributes including ineffective/negative maternal and paternal discipline.³⁵ The likelihood that parental psychopathology and family dysfunction act as contributors to childhood aggressive behavior suggests that dedicated parental or family therapy may be required in many cases as an adjunct to parent

management training.⁷¹ Parent monitoring of peers and peer activities may also be beneficial as relationships with deviant peers can influence later development of antisocial behavior.⁷²

Efficacy: Pharmacotherapy

There is strong evidence for the efficacy of medication, especially stimulants, in ADHD treatment, and stimulants represent appropriate first-line treatment alone or in combination with behavior therapy.⁶⁶ Unlike the evidence for behavioral treatments, however, there is a lack of data regarding the efficacy of medications on symptoms of ODD and CD in children without ADHD.⁷⁰ A summary of individual studies on the efficacy of pharmacotherapy to treat oppositional symptoms in patients with ADHD with or without ODD/CD is provided in Table 3.^{27,60,62,73–85}

With respect to the direct effects of stimulant medications on oppositional symptoms, a 2002 meta-analysis of 28 studies of stimulant medications concluded that these medications have a significant, reproducible, and robust effect on oppositional symptoms, which were included in the assessment of overt aggression for the purposes of this analysis. As shown in Table 3, the weighted effect size was 0.84 for overt aggression, which is comparable in magnitude to the effect sizes routinely reported for stimulants in treating core ADHD symptoms.⁸³

The studies of methylphenidate summarized in Table 3 found it to be effective in the treatment of oppositional symptoms. Those studies that reported safety data for methylphenidate primarily reported anorexia and stomachache as common side effects.^{73,75} By comparison, evidence regarding the efficacy of atomoxetine and clonidine are mixed. Of the 4 studies (or subset analyses) reported in Table 3, 2 found efficacy of atomoxetine in the treatment of oppositional symptoms in children with both ADHD and ODD at particular doses⁶⁰ or time points.⁷⁷ One study found efficacy of atomoxetine only in the group of subjects with both ADHD and ODD, but not in the group with ADHD alone.⁶⁰ Another study that found efficacy of atomoxetine in the treatment of oppositional symptoms examined a population of subjects with ADHD, approximately 38% of whom also had comorbid ODD.⁶² The subset analysis of children with ADHD and ODD from 2 studies found no significant effects of atomoxetine on oppositional symptoms.⁸¹ Furthermore, 2 studies found that the efficacy of atomoxetine in the treatment of oppositional symptoms was highly correlated with the efficacy in ADHD symptoms, such that when the impact of atomoxetine on ADHD symptoms was controlled for, the impact of atomoxetine on oppositional symptoms was not significant.^{60,77} The 2 meta-analyses also reported conflicting results, with 1 finding efficacy of atomoxetine in subjects with comorbid ADHD and ODD,⁸² and the other finding no significant effect of atomoxetine in subjects with ADHD, a substantial proportion of whom had comorbid ODD.⁸⁴ Among those studies that reported safety data for atom-

Table 3. Efficacy of Pharmacotherapy in the Treatment of Oppositional Symptoms

References	Treatment	Methodology	Sample Size	Outcome Measure(s)	Efficacy
Connor et al ⁸³	Stimulant medications (MPH, amphetamine, and pemoline)	Meta-analysis of 28 studies	683 patients with ADHD	Clinician, parent, and teacher ratings of overt aggression	Weighted effect size of 0.84 for overt aggression
Gorman et al ²⁷	MPH (≤ 0.4 mg/kg bid + 0.2 mg/kg in the late afternoon)	Double-blind crossover study	N = 75 children (41 with ADHD and 34 community controls)	IOWA-CRS	Significant improvement in aggression/oppositionality in patients with the combined subtype of ADHD (effect size [η_p^2] = 0.42; $p < .001$) No significant improvement in patients with the inattentive subtype of ADHD ^a
Schachar et al ⁷³	MPH (target dose of 0.7 mg/kg twice daily)	Randomized, placebo-controlled study	N = 91 children with ADHD	TIP and IOWA-CRS	Significantly greater improvements with MPH than placebo on teacher ratings of oppositional behavior ($p < .005$) and aggression ($p < .001$), as assessed by the TIP and IOWA-CRS, respectively No significant benefit of MPH treatment over placebo on parental ratings of oppositional symptoms or aggression
van der Oord et al ⁷⁴	Multimodal behavior therapy as an adjunct to MPH treatment (max 0.9 mg/kg)	Pseudo-randomized, multiple-blind placebo-controlled, crossover medication design	N = 50 children with ADHD	Parent- and teacher-rated ODD and CD subscales of the DBDRS	Significant improvement in the ODD subscale of the DBDRS (Effect size [η_p^2] = 0.25 for parent ratings, η_p^2 = 0.31 for teacher ratings; $p < .01$ for both) Significant improvement in the CD subscale of the DBDRS (η_p^2 = 0.24 for parent ratings, η_p^2 = 0.26 for teacher ratings; $p < .01$ for both) No significant differences were seen between patients receiving MPH alone and patients also receiving multimodal behavioral therapy
Pelham et al ⁷⁵	MPH (10 or 17.5 mg bid, MAS (7.5 or 12.5 mg bid)	Within-subject, placebo-controlled, crossover design	N = 25 children with ADHD, 21 with comorbid ODD or CD	Point system Classroom observational measures Counselor-, teacher-, and parent-rated IOWA-CRS	Significantly greater improvements with stimulants than placebo for following rules, noncompliance, and conduct problems as measured by a point system, disruptive behaviors as measured by classroom observation Significantly greater improvements with stimulants than placebo on counselor-, teacher-, and parent-rated IOWA-CRS
Gadow et al ⁷⁶	MPH (0.1, 0.3, and 0.5 mg/kg)	Randomized, double-blind, placebo-controlled, crossover study	N = 31 children with ADHD + ODD and coexisting chronic multiple tic disorder or Tourette disorder	Teacher-rated oppositional/defiant subscale of the IOWA-CRS MOMS checklist Teacher-rated ACRS Factor 2 Parent-rated ACRS Factor 2	Significantly greater improvements with MPH than placebo in oppositional symptoms, as measured by the oppositional/defiant subscale of the IOWA-CRS (Effect size [Cohen's d] = 0.75; p = .0008), the MOMS checklist (Cohen's d = 0.48; p = .0046), and the teacher-rated ACRS Factor 2 (Cohen's d = 0.48; p = .0016) but not the parent-rated ACRS Factor 2

(Table continues)

Table 3. Continued

References	Treatment	Methodology	Sample Size	Outcome Measure(s)	Efficacy
Spencer et al ⁸⁵	MAS XR (10, 20, 30, and 40 mg/d)	Randomized, double-blind, placebo-controlled, parallel-group, forced-dose-escalation study	N = 308 children and adolescents with ODD (62 with ODD without ADHD)	SNAP-IV ODD subscale (parent rating) SNAP-IV ODD subscale (teacher rating)	Significantly greater improvements on the ODD subscale of the SNAP-IV parent rating for the MAS XR 30-mg/d dose group ($p < .005$) but not the 10-, 20-, or 40-mg/d groups (ITT population) Significantly greater improvements on the ODD subscale of the SNAP-IV teacher rating for the 10-, 20-, and 30-mg/d dose groups ($p < .005$ for all) but not the 40-mg/d group (ITT population) No significant improvements in the SNAP-IV parent rating scores were seen in any dose group in the group of subjects with ODD without ADHD
Biederman et al ⁸⁴	Atomoxetine (up to 1.8 mg/kg/d) ^b	Meta-analysis of 3 studies	N = 512 school-aged children with ADHD; 158 had been diagnosed with comorbid ODD	CPRS-R:S oppositional subscale score	Although atomoxetine treatment produced a 3-fold greater numerical reduction from baseline in patients with ADHD+ODD than in those with ADHD alone, these reductions were not significantly different from those with placebo
Newcom et al ⁶⁰	Atomoxetine (0.5, 1.2, and 1.8 mg/kg/d) ^b	Randomized, double-blind, placebo-controlled study	N = 297 children with ADHD, 115 with comorbid ODD	CPRS-R:S oppositional subscale	Atomoxetine reduced symptoms of oppositionality (as measured by the CPRS-R:S oppositional subscale) in children with ADHD+ODD at 0.5 ($p = .040$) and 1.8 ^b ($p = .027$) mg/kg/d although not at 1.2 mg/kg/d Effect sizes in children with ADHD+ODD ranged from 0.39 to 0.68
Bangs et al ⁷⁷	Atomoxetine (1.2 mg/kg)	Randomized, double-blind placebo-controlled study	N = 226 children with ADHD and comorbid ODD	Investigator-rated ODD subscale of SNAP-IV	Significantly greater change from baseline on the ODD subscale of the SNAP-IV with atomoxetine than placebo in the overall repeated-measures analysis ($p = .01$) Significant effect of atomoxetine treatment at weeks 2 and 5, although not at week 8 No significant difference between the atomoxetine-treated and placebo groups as determined by a last-observation-carried-forward analysis
Michelson et al ⁶²	Atomoxetine (0.5, 1.2, and 1.8 mg/kg/d) ^b	Randomized, placebo-controlled study	N = 297 children with ADHD (comorbid ODD was present in approximately 38% of subjects)	CPRS-R oppositional subscale	Significantly greater improvements with atomoxetine than placebo in oppositional symptoms ($p < .05$)

(Table continues)

Table 3. Continued

References	Treatment	Methodology	Sample Size	Outcome Measure(s)	Efficacy
Kaplan et al ⁸¹	Atomoxetine (up to 2 mg/kg/d or 90 mg/d, whichever was lower) ^b	Subset analysis of 2 randomized, double-blind, placebo-controlled studies	N = 98 children with ADHD and comorbid ODD	CPRS-R:S oppositional subscale	No significant difference between the atomoxetine-treated and placebo groups
Cheng et al ⁸²	Atomoxetine (various doses)	Meta-analysis and meta-regression analysis	9 studies, n = 137 subjects treated with atomoxetine, n = 76 subjects treated with placebo	CPRS-R:S oppositional subscale	Significant benefits of atomoxetine treatment compared with placebo ($p < .05$) in subjects with ADHD and ODD
Hunt et al ⁷⁸	Clonidine (~4–5 µg/kg/d)	Randomized, double-blind study	N = 10 children with ADHD	Conners' Teachers Behavior Rating Conners' Parents Behavior Rating	Significant improvement from baseline with clonidine compared with placebo in teacher-rated conduct problems ($p = .04$) Significant improvement from baseline with clonidine compared with placebo in parent-rated conduct problems ($p = .01$)
Connor et al ⁷⁹	Clonidine monotherapy (≤ 0.3 mg/d), MPH monotherapy (≤ 40 mg/d), or clonidine + MPH	Randomized, double-blind, parallel-group study	N = 24 male children with ADHD+ODD or ADHD+CD	Parent- and teacher-rated DBS	Significant reductions from baseline in parent-rated ODD ($p = .00008$), and CD symptom severity ($p = .00009$) and in teacher-rated and ODD symptom severity ($p = .003$) with clonidine alone, MPH alone, and clonidine + MPH
Palumbo et al ⁸⁰	Clonidine monotherapy (≤ 0.6 mg/d), MPH monotherapy (≤ 60 mg/d), or clonidine + MPH	Randomized, double-blind, placebo-controlled study	N = 122 children with ADHD (~50% with comorbid ODD)	Conners' ASQ for Parents and for Teachers Iowa Conners impulsive/oppositional/defiant subscales Direct classroom observations of disruptive behaviors	No significant treatment effect with clonidine, whether administered alone or in combination with MPH, as measured by the Conners ASQ for Teachers No significant effect with clonidine on the Iowa Conners impulsive/oppositional or oppositional/defiant subscales or direct classroom observations of disruptive behaviors Significant effect of clonidine on the Conners ASQ for Parents ($p = .0003$)

MPH, methylphenidate; ADHD, attention-deficit/hyperactivity disorder; IOWA-CTS, IOWA Conners' Rating Scale; IOWA-CTRS, IOWA Conners Teacher's Rating Scale; TIP, telephone interview probe; ODD, oppositional defiant disorder; CD, conduct disorder; DBDRS, Disruptive Behavior Disorder Rating Scale; MOMS, Mothers' Objective Method for Subgrouping; ACRS, Abbreviated Conners' Rating Scale; CPRS-R, Conners' Parent Rating Scale—Revised; CPRS-R:S, Conners' Parent Rating Scale—Revised; Short Form; SNAP-IV, Swanson, Nolan, and Pelham Rating Scale; DBS, Disruptive Behavior Scale; Conners' ASQ, Conners' Abbreviated Symptoms Questionnaire; MAS, mixed amphetamine salts; MAS XR, mixed amphetamine salts extended release; ITT, intention-to-treat. ^aThe lack of significance in this group may be partially attributable to "floor effects" (i.e., a decreased potential for improvement based on lower baseline symptom severity). ^bDoses of atomoxetine greater than 1.4 mg/kg/d for children and adolescents up to 70 kg or 100 mg/d for children and adolescents greater than 70 kg and adults are greater than the recommended upper dose for treatment of ADHD as listed in the US Food and Drug Administration prescribing information for atomoxetine.

oxetine, decreased appetite and somnolence were most commonly mentioned.^{62,77,81,82} The α_2 -adrenergic agonist clonidine has not been approved for the treatment of ADHD, although it has been investigated as an agent that may be useful in patients with insufficient response to stimulants. There are fewer studies that evaluated clonidine in the treatment of oppositional symptoms, although 2 small studies found efficacy in the treatment of oppositional symptoms.^{78,79} However, a larger study found mixed results, depending on the measure.⁸⁰ Sleepiness, somnolence, and fatigue were commonly reported with clonidine treatment.^{78,80,86}

The extended-release formulation of guanfacine, another α_2 -adrenergic agonist, has recently been approved by the Food and Drug Administration for the treatment of ADHD in children and adolescents aged 6 to 17 years.⁸⁷ The most common treatment emergent adverse events seen in 2 short-term pivotal studies of extended-release formulation of guanfacine included somnolence, fatigue, and upper abdominal pain.^{88,89} Children with preexisting cardiovascular conditions or syncope are not candidates for extended-release formulation of guanfacine or other alpha-agonist medications. A study of extended-release formulation of guanfacine in the treatment of oppositional symptoms in children aged 6 to 12 years with a diagnosis of ADHD and the presence of oppositional symptoms has been completed and the results have been submitted for publication.

Although evidence regarding the efficacy of medication in the treatment of oppositional symptoms in ADHD is mixed, the potential response to pharmacotherapy makes clinical evaluation of these symptoms relevant for effective treatment. The possibility of effective treatment for comorbid oppositional symptoms should encourage health care providers to carefully screen their patients with ADHD for these symptoms with the knowledge that potentially effective behavioral and pharmacologic interventions are available.

DISCUSSION

Current Controversies

Current controversies include whether oppositional defiant disorder (ODD) in the context of Attention-deficit/hyperactivity disorder (ADHD) is a precursor of later conduct disorder (CD), whether combined psychosocial and medication therapies have greater effects in patients with ADHD with comorbid ODD or CD than does monotherapy alone, and whether current pharmacotherapy is effective for oppositional symptoms in the context of ADHD treatment. The relationships among ADHD, oppositional symptoms, and diagnosed ODD and CD are complex, and these relationships are, to some extent, affected by changing definitions and criterion symptoms. For example, it remains unresolved whether oppositional symptoms in association with ADHD represent a truly distinct diagnosis (ODD or CD) or should be considered subsyndromal (i.e., a component of ADHD) to ADHD itself.^{84,90} Moreover, at various times, ODD has

been considered both a subsyndrome and a precursor of CD. Within the context of ADHD, the temporal relationship between ODD and CD remains unclear. One longitudinal study suggested that 2 subtypes of ODD may be comorbid with ADHD: in one, ODD appears to be prodromal to CD, with a high likelihood of subsequent CD; and in the other, ODD appears to be subsyndromal to CD and is unlikely to progress to subsequent CD.²⁵ Further longitudinal studies should help clarify these relationships and the extent to which they are affected by evolving definitions, and ultimately help guide treatment approaches for oppositional symptoms associated with ADHD. A critical corollary issue for such studies to address is whether early treatment of a putative prodromal ODD in combination with ADHD can reduce the risk of progression to CD.

The ambiguity and unresolved issues surrounding the relationship between ADHD, ODD, CD, and aggressive symptoms are a concern when interpreting existing data. The common practice of considering subjects with ADHD with comorbid ODD or ADHD with comorbid CD together limits the conclusions that can be drawn from those studies. Until a method of defining these disorders in a valid and useful way is identified, these concerns will remain.

A further limitation to determining the clinical impact of the existing data is the lack of studies that differentiate between the subtypes of ADHD. Despite a higher prevalence of ODD/CD in patients with the combined ADHD subtype than in those with the inattentive subtype, most studies do not consider individual subtypes when analyzing data. Only one study, which is described in Table 3, examined efficacy of methylphenidate on oppositional symptoms by ADHD subtypes. Significant improvement in aggression/oppositionality was seen in patients with the combined, although not the inattentive, subtype of ADHD. However, this may have been due to the low baseline levels of aggression/oppositionality seen in this group.²⁷

As suggested by the evidence cited here, the optimal approach to treatment of ADHD with comorbid oppositional symptoms remains unclear. Although behavior therapy, anchored by parent management training, is a mainstay of ADHD treatment, it remains to be resolved whether combination psychosocial therapy plus pharmacotherapy are more effective in ADHD with comorbid oppositional symptoms than are individual therapies and whether ADHD and ODD/CD should be targeted by agents that treat both together or each individually. This question is made more complex by the fact that outcome measures differ in their sensitivity to medication versus behavioral treatments (stimulants have large effects on ADHD symptom severity, whereas behavioral treatments may exert greater effects on measures of functional impairment that reflect the targets of the treatment). Given the current knowledge of treatment options for ODD, there is general consensus that behavioral treatment is optimal for patients

with ODD alone, and that medication is recommended only as a potential adjunctive therapy.^{7,15}

Although the evidence cited here remains unclear with respect to the impact of medication on “pure” ODD (in the absence of ADHD), there is some support for the efficacy of medication for oppositional symptoms in the presence of ADHD, although higher doses of stimulants or atomoxetine than those used in the treatment of ADHD alone may be necessary to achieve efficacy. Health care providers should therefore consider multimodal therapy for patients with ADHD+ODD. Because psychosocial treatments are not consistently standardized or available to all patients, it is important for physicians to find providers who offer evidence-based psychosocial treatment, generally including behavioral parent training and behavioral family therapy. This evidence needs to be clarified in additional clinical studies, but, in the meantime, the potential to improve oppositional symptoms with medication reinforces the importance of assessing the presence of these symptoms in any patient with ADHD.

CONCLUSIONS

Despite the evolving knowledge base described herein, several aspects of oppositional symptoms in the context of attention-deficit/hyperactivity disorder (ADHD) are clear. Oppositional symptoms, whether subsyndromal or in the form of comorbid oppositional defiant disorder and/or conduct disorder, are a key consideration in the screening, diagnosis, and management of ADHD symptoms; their evaluation during the course of treatment is as important as that of core ADHD symptoms. The presence of comorbid oppositional symptoms in a patient with ADHD implies the need to accommodate a greater degree of complexity with respect to ADHD management relative to “pure” ADHD without comorbid oppositional symptoms. Despite this incremental complexity, clinical studies suggest that ADHD symptoms, as well as oppositional symptoms, are amenable to effective treatment, although all pharmacologic treatments have been associated with adverse events. Further research should seek to clarify whether oppositional symptoms are subsyndromal to ADHD or if they require a separate diagnosis. Future studies should also explore the possibility that higher dosages of stimulants may be required to produce the same level of symptomatic improvement in patients with ADHD complicated by oppositional symptoms as in patients with ADHD without oppositional symptoms. In addition, further research with noradrenergic agents such as atomoxetine and the α_2 -adrenoceptor agonists should help clarify their role in the treatment of complex ADHD.

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REFERENCES

1. American Psychiatric Association. Attention-deficit and disruptive behavior disorders. In: *Diagnostic and Statistical Manual of Mental Disorders DSM-IV-TR, Fourth Edition. Text Revision*. Washington, DC: American Psychiatric Association; 2000:85–93.
2. Reiff MI, Stein MT. Attention-deficit/hyperactivity disorder evaluation and diagnosis. A practical approach in office practice. *Pediatr Clin North Am*. 2003;50:1019–1048.
3. Biederman J, Faraone SV. Attention-deficit hyperactivity disorder. *Lancet*. 2005;366:237–248.
4. Biederman J. Attention-deficit/hyperactivity disorder: a selective overview. *Biol Psychiatry*. 2005;57:1215–1220.
5. Centers for Disease Control and Prevention. Mental health in the United States: prevalence of diagnosis and medication treatment for attention-deficit/hyperactivity disorder—United States, 2003. *MMWR Morb Mortal Wkly Rep*. 2005;54:842–847.
6. Piffner LJ. Comorbidity as an organizing principle. In: Keith McBurnett, Linda Piffner, eds. *Attention Deficit Hyperactivity Disorder: Concepts, Controversies, New Directions*. New York, NY: Informa HealthCare; 2007:51–61.
7. Steiner H, Remsing L; Work Group on Quality Issues. Practice parameter for the assessment and treatment of children and adolescents with oppositional defiant disorder. *J Am Acad Child Adolesc Psychiatry*. 2007;46:126–141.
8. Lahey BB, Applegate B, Waldman ID, Loft JD, Hankin BL, Rick J. The structure of child and adolescent psychopathology: generating new hypotheses. *J Abnorm Psychol*. 2004;113:358–385.
9. Newcorn JH, Halperin JM. Attention-deficit disorders with oppositionality and aggression. In: Brown TE, ed. *Attention-Deficit Disorders and Comorbidities in Children, Adolescents, and Adults*. Washington, DC: American Psychiatric Press, Inc.; 2000:171–207.
10. Connor DF, Doerfler LA. ADHD with comorbid oppositional defiant disorder or conduct disorder: discrete or nondistinct disruptive behavior disorders? *J Atten Disord*. 2008;12:126–134.
11. Lahey BB, Loeber R, Burke J, Rathouz PJ, McBurnett K. Waxing and waning in concert: dynamic comorbidity of conduct disorder with other disruptive and emotional problems over 7 years among clinic-referred boys. *J Abnorm Psychol*. 2002;111:556–567.
12. Burke JD, Loeber R, Lahey BB, Rathouz PJ. Developmental transitions among affective and behavioral disorders in adolescent boys. *J Child Psychol Psychiatry*. 2005;46:1200–1210.
13. Pliszka SR. Comorbidity of attention-deficit/hyperactivity disorder with psychiatric disorder: an overview. *J Clin Psychiatry*. 1998;59(suppl 7):50–58.
14. Lahey BB, Van Hulle CA, Rathouz PJ, Rodgers JL, D’Onofrio BM, Waldman ID. Are oppositional-defiant and hyperactive-inattentive symptoms developmental precursors to conduct problems in late childhood?: genetic and environmental links. *J Abnorm Child Psychol*. 2009;37:45–58.
15. Burke JD, Loeber R, Birmaher B. Oppositional defiant disorder and conduct disorder: a review of the past 10 years, part II. *J Am Acad Child Adolesc Psychiatry*. 2002;41:1275–1293.
16. Loeber R, Burke JD, Lahey BB, Winters A, Zera M. Oppositional defiant and conduct disorder: a review of the past 10 years, part I. *J Am Acad Child Adolesc Psychiatry*. 2000;39:1468–1484.
17. Piffner LJ, McBurnett K, Lahey BB, et al. Association of parental psychopathology to the comorbid disorders of boys with attention deficit-hyperactivity disorder. *J Consult Clin Psychol*. 1999;67:881–893.
18. Hartup WW. The development of aggression: where do we stand? In: Tremblay RE, Hartup WW, Archer J, eds.

Developmental Origins of Aggression. New York, NY: The Guilford Press; 2005:3–22.

19. Odgers CL, Moffitt TE, Broadbent JM, et al. Female and male antisocial trajectories: from childhood origins to adult outcomes. *Dev Psychopathol*. 2008;20:673–716.
20. Moffitt TE. Adolescence-limited and life-course-persistent antisocial behavior: a developmental taxonomy. *Psychol Rev*. 1993;100:674–701.
21. Campbell SB, Spieker S, Burchinal M, Poe MD; NICHD Early Child Care Research Network. Trajectories of aggression from toddlerhood to age 9 predict academic and social functioning through age 12. *J Child Psychol Psychiatry*. 2006;47:791–800.
22. Côté SM, Vaillancourt T, LeBlanc JC, Nagin DS, Tremblay RE. The development of physical aggression from toddlerhood to pre-adolescence: a nation wide longitudinal study of Canadian children. *J Abnorm Child Psychol*. 2006;34:71–85.
23. Tremblay RE, Nagin DS, Seguin JR, et al. Physical aggression during early childhood: trajectories and predictors. *Pediatrics*. 2004;114:e43–e50.
24. Loeber R, Burke J, Pardini DA. Perspectives on oppositional defiant disorder, conduct disorder, and psychopathic features. *J Child Psychol Psychiatry*. 2009;50:133–142.
25. Biederman J, Faraone SV, Milberger S, et al. Is childhood oppositional defiant disorder a precursor to adolescent conduct disorder? Findings from a four-year follow-up study of children with ADHD. *J Am Acad Child Adolesc Psychiatry*. 1996;35:1193–1204.
26. Pliszka SR, Carlson CL, Swanson JM. Disruptive behavior disorders and substance abuse. In: *ADHD With Comorbid Disorders: Clinical Assessment and Management*. New York, NY: The Guilford Press; 1999:89–112.
27. Gorman EB, Klorman R, Thatcher JE, Borgstedt AD. Effects of methylphenidate on subtypes of attention-deficit/hyperactivity disorder. *J Am Acad Child Adolesc Psychiatry*. 2006;45:808–816.
28. Loeber R, Green SM, Lahey BB, Frick PJ, McBurnett K. Findings on disruptive behavior disorders from the first decade of the Developmental Trends Study. *Clin Child Fam Psychol Rev*. 2000;3:37–60.
29. Molina BS, Pelham WE Jr. Childhood predictors of adolescent substance use in a longitudinal study of children with ADHD. *J Abnorm Psychol*. 2003;112:497–507.
30. Satterfield J, Swanson J, Schell A, Lee F. Prediction of antisocial behavior in attention-deficit hyperactivity disorder boys from aggression/defiance scores. *J Am Acad Child Adolesc Psychiatry*. 1994;33:185–190.
31. Barkley RA, Guevremont DC, Anastopoulos AD, DuPaul GJ, Shelton TL. Driving-related risks and outcomes of attention deficit hyperactivity disorder in adolescents and young adults: a 3- to 5-year follow-up survey. *Pediatrics*. 1993;92:212–218.
32. Harpold T, Biederman J, Gignac M, et al. Is oppositional defiant disorder a meaningful diagnosis in adults? Results from a large sample of adults with ADHD. *J Nerv Ment Dis*. 2007;195:601–605.
33. Pfiffner LJ, McBurnett K, Rathouz PJ. Father absence and familial antisocial characteristics. *J Abnorm Child Psychol*. 2001;29:357–367.
34. Werner EE. High-risk children in young adulthood: a longitudinal study from birth to 32 years. *Am J Orthopsychiatry*. 1989;59:72–81.
35. Pfiffner LJ, McBurnett K, Rathouz PJ, Judice S. Family correlates of oppositional and conduct disorders in children with attention deficit/hyperactivity disorder. *J Abnorm Child Psychol*. 2005;33:551–563.
36. Mikami AY, Pfiffner LJ. Sibling relationships among children with ADHD. *J Atten Disord*. 2008;11:482–492.
37. Klassen AF, Miller A, Fine S. Health-related quality of life in children and adolescents who have a diagnosis of attention-deficit/hyperactivity disorder. *Pediatrics*. 2004;114:e541–e547.
38. Pliszka S; AACAP Work Group on Quality Issues. Practice parameter for the assessment and treatment of children and adolescents with attention-deficit/hyperactivity disorder. *J Am Acad Child Adolesc Psychiatry*. 2007;46:894–921.
39. Lahey BB, Applegate B, McBurnett K, et al. DSM-IV field trials for attention deficit hyperactivity disorder in children and adolescents. *Am J Psychiatry*. 1994;151:1673–1685.
40. McBurnett K, Pfiffner LJ, Willcutt E, et al. Experimental cross-validation of DSM-IV types of attention-deficit/hyperactivity disorder. *J Am Acad Child Adolesc Psychiatry*. 1999;38:17–24.
41. Lahey BB, Applegate B, Barkley RA, et al. DSM-IV field trials for oppositional defiant disorder and conduct disorder in children and adolescents. *Am J Psychiatry*. 1994;151:1163–1171.
42. Hart EL, Lahey BB, Loeber R, Hanson KS. Criterion validity of informants in the diagnosis of disruptive behavior disorders in children: a preliminary study. *J Consult Clin Psychol*. 1994;62:410–414.
43. Conners CK. *Conners' Rating Scales—Revised Technical Manual*. North Tonawanda, NY: Multi-Health Systems Inc; 2001.
44. Swanson JM. The SNAP-IV Teacher and Parent Rating Scale. Available at: <http://www.adhd.net/snap-iv-form.pdf>. Accessed January 26, 2010.
45. Scoring Instructions for the SNAP-IV-C Rating Scale. Available at: <http://www.adhd.net/snap-iv-instructions.pdf>. Accessed January 26, 2010.
46. Conners CK. *Conners Manual*. 3rd ed. Ontario, Canada: Multi-Health Systems Inc.; 2008.
47. Gadow KD, Sprafkin J. *Child Symptom Inventory: Ages 6 to 12 Years (CSI-4)*. Lutz, FL: Psychological Assessment Resources, Inc.; 2000.
48. Vanderbilt ADHD Diagnostic Parent Rating Scale. Available at: http://imaging.cmpmedica.com/CME/pt/PDF/VADPRS_Quest.pdf. Accessed January 26, 2010.
49. Vanderbilt ADHD Diagnostic Parent Rating Scale Scoring Instructions. Available at: http://imaging.cmpmedica.com/CME/pt/PDF/VADPRS_Instructs.pdf. Accessed January 26, 2010.
50. Gadow KD, Sprafkin J, Carlson GA, et al. A DSM-IV-referenced, adolescent self-report rating scale. *J Am Acad Child Adolesc Psychiatry*. 2002;41:671–679.
51. Mathias CW, Stanford MS, Marsh DM, et al. Characterizing aggressive behavior with the Impulsive/Premeditated Aggression Scale among adolescents with conduct disorder. *Psychiatry Res*. 2007;151:231–242.
52. Burney DM, Kromrey J. Initial development and score validation of the Adolescent Anger Rating Scale. *Educ Psychol Meas*. 2001;61:446–460.
53. Nelson WMI, Finch AJ Jr. *Children's Inventory of Anger (ChIA) Manual*. Los Angeles, CA: Western Psychological Services, 2000.
54. Achenbach System of Empirically Based Assessment. *Youth Self Report/11–18*. Burlington, VT: Achenbach System of Empirically Based Assessment; 2009.
55. Findling RL, Connor DF, Wigal T, Eagan C, Onofrey MN. A linguistic analysis of in-office dialogue among psychiatrists, parents and child and adolescent patients with ADHD. *J Atten Disord*. 2009;13:78–86.
56. Jensen PS, Hinshaw SP, Kraemer HC, et al. ADHD comorbidity findings from the MTA study: comparing comorbid subgroups. *J Am Acad Child Adolesc Psychiatry*. 2001;40:147–158.
57. Klein RG, Abikoff H, Hechtman L, Weiss G. Design and rationale of controlled study of long-term methylphenidate and multimodal psychosocial treatment in children with ADHD. *J Am Acad Child Adolesc Psychiatry*. 2004;43:792–801.
58. Abikoff H, Hechtman L, Klein RG, et al. Symptomatic improvement in children with ADHD treated with long-term methylphenidate and multimodal psychosocial treatment. *J Am Acad Child Adolesc Psychiatry*. 2004;43:802–811.
59. Spencer TJ, Abikoff HB, Connor DF, et al. Efficacy and safety of mixed amphetamine salts extended release (adderall XR) in the management of oppositional defiant disorder with or without

- comorbid attention-deficit/hyperactivity disorder in school-aged children and adolescents: a 4-week, multicenter, randomized, double-blind, parallel-group, placebo-controlled, forced-dose-escalation study. *Clin Ther*. 2006;28:402–418.
60. Newcorn JH, Spencer TJ, Biederman J, Milton DR, Michelson D. Atomoxetine treatment in children and adolescents with attention-deficit/hyperactivity disorder and comorbid oppositional defiant disorder. *J Am Acad Child Adolesc Psychiatry*. 2005;44:240–248.
 61. *Strattera* [package insert]. Indianapolis, IN: Eli Lilly and Company; 2009.
 62. Michelson D, Faries D, Wernicke J, et al; Atomoxetine ADHD Study Group. Atomoxetine in the treatment of children and adolescents with attention-deficit/hyperactivity disorder: a randomized, placebo-controlled, dose-response study. *Pediatrics*. 2001;108:E83.
 63. Connor DF, Fletcher KE, Swanson JM. A meta-analysis of clonidine for symptoms of attention-deficit hyperactivity disorder. *J Am Acad Child Adolesc Psychiatry*. 1999;38:1551–1559.
 64. Thiruchelvam D, Charach A, Schachar RJ. Moderators and mediators of long-term adherence to stimulant treatment in children with ADHD. *J Am Acad Child Adolesc Psychiatry*. 2001;40:922–928.
 65. Evans SW, Schultz BK, Sadler JM. Safety and efficacy of psychosocial interventions used to treat children with attention-deficit/hyperactivity disorder. *Pediatr Ann*. 2008;37:52–59.
 66. American Academy of Pediatrics. Subcommittee on Attention-Deficit/Hyperactivity Disorder and Committee on Quality Improvement. Clinical practice guideline: treatment of the school-aged child with attention-deficit/hyperactivity disorder. *Pediatrics*. 2001;108:1033–1044.
 67. Pelham WEJ, Fabiano GA. Behavior modification. *Child Adolesc Psychiatr Clin North Am*. 2000;9:671–688.
 68. Barkley RA. *Defiant Children: A Clinician's Manual for Assessment and Parent Training*. 2nd ed. New York, NY: Guilford Press; 1997.
 69. Young S, Myanthy AJ. Practitioner review: non-pharmacological treatments for ADHD: a lifespan approach. *J Child Psychol Psychiatry*. 2009;51:116–133.
 70. Eyberg SM, Nelson MM, Boggs SR. Evidence-based psychosocial treatments for children and adolescents with disruptive behavior. *J Clin Child Adolesc Psychol*. 2008;37:215–237.
 71. Pliszka SR. Psychiatric comorbidities in children with attention deficit hyperactivity disorder: implications for management. *Paediatr Drugs*. 2003;5:741–750.
 72. Patterson GR, Dishion TJ, Yoerger K. Adolescent growth in new forms of problem behavior: macro- and micro-peer dynamics. *Prev Sci*. 2000;1:3–13.
 73. Schachar RJ, Tannock R, Cunningham C, Corkum PV. Behavioral, situational, and temporal effects of treatment of ADHD with methylphenidate. *J Am Acad Child Adolesc Psychiatry*. 1997;36:754–763.
 74. van der Oord S, Prins PJ, Oosterlaan J, Emmelkamp PM. Does brief, clinically based, intensive multimodal behavior therapy enhance the effects of methylphenidate in children with ADHD? *Eur Child Adolesc Psychiatry*. 2007;16:48–57.
 75. Pelham WE, Aronoff HR, Midlam JK, et al. A comparison of Ritalin® and Adderall®: efficacy and time-course in children with attention-deficit/hyperactivity disorder. *Pediatrics*. 1999;103:e43.
 76. Gadow KD, Nolan EE, Sverd J, Sprafkin J, Schneider J. Methylphenidate in children with oppositional defiant disorder and both comorbid chronic multiple tic disorder and ADHD. *J Child Neurol*. 2008;23:981–990.
 77. Bangs ME, Hazell P, Danckaerts M, et al; Atomoxetine ADHD/ODD Study Group. Atomoxetine for the treatment of attention-deficit/hyperactivity disorder and oppositional defiant disorder. *Pediatrics*. 2008;121:e314–e320.
 78. Hunt RD, Minderaa RB, Cohen DJ. Clonidine benefits children with attention deficit disorder and hyperactivity: report of a double-blind placebo-crossover therapeutic trial. *J Am Acad Child Psychiatry*. 1985;24:617–629.
 79. Connor DF, Barkley RA, Davis HT. A pilot study of methylphenidate, clonidine, or the combination in ADHD comorbid with aggressive oppositional defiant or conduct disorder. *Clin Pediatr*. 2000;39:15–25.
 80. Palumbo DR, Sallee FR, Pelham WE Jr, Bukstein OG, Daviss WB, McDermott MP. Clonidine for attention-deficit/hyperactivity disorder: I. Efficacy and tolerability outcomes. *J Am Acad Child Adolesc Psychiatry*. 2008;47:180–188.
 81. Kaplan S, Heiligenstein J, West S, et al. Efficacy and safety of atomoxetine in childhood attention-deficit/hyperactivity disorder with comorbid oppositional defiant disorder. *J Atten Disord*. 2004;8:45–52.
 82. Cheng JY, Chen RY, Ko JS, Ng EM. Efficacy and safety of atomoxetine for attention-deficit/hyperactivity disorder in children and adolescents-meta-analysis and meta-regression analysis. *Psychopharmacology (Berl)*. 2007;194:197–209.
 83. Connor DF, Glatt SJ, Lopez ID, Jackson D, Melloni RH Jr. Psychopharmacology and aggression. I: a meta-analysis of stimulant effects on overt/covert aggression-related behaviors in ADHD. *J Am Acad Child Adolesc Psychiatry*. 2002;41:253–261.
 84. Biederman J, Spencer TJ, Newcorn JH, et al. Effect of comorbid symptoms of oppositional defiant disorder on responses to atomoxetine in children with ADHD: a meta-analysis of controlled clinical trial data. *Psychopharmacology (Berl)*. 2007;190:31–41.
 85. Spencer TJ, Wilens TE, Biederman J, Weisler RH, Read SC, Pratt R. Efficacy and safety of mixed amphetamine salts extended release (Adderall XR) in the management of attention-deficit/hyperactivity disorder in adolescent patients: a 4-week, randomized, double-blind, placebo-controlled, parallel-group study. *Clin Ther*. 2006;28:266–279.
 86. Daviss WB, Patel NC, Robb AS, et al. Clonidine for attention-deficit/hyperactivity disorder: II. ECG changes and adverse events analysis. *J Am Acad Child Adolesc Psychiatry*. 2008;47:189–198.
 87. *Intuniv* [package insert]. Wayne, PA: Shire Pharmaceuticals Inc.; 2009.
 88. Biederman J, Melmed RD, Patel A, et al; SPD503 Study Group. A randomized, double-blind, placebo-controlled study of guanfacine extended release in children and adolescents with attention-deficit/hyperactivity disorder. *Pediatrics*. 2008;121:e73–e84.
 89. Sallee F, McGough J, Wigal T, Donahue J, Lyne A, Biederman J; SPD503 Study Group. Guanfacine extended release in children and adolescents with attention-deficit/hyperactivity disorder: a placebo-controlled trial. *J Am Acad Child Adolesc Psychiatry*. 2009;48:155–165.
 90. Lahey BB, Rathouz PJ, Van Hulle C, et al. Testing structural models of DSM-IV symptoms of common forms of child and adolescent psychopathology. *J Abnorm Child Psychol*. 2008;36:187–206.